

When Does an Image Edit Become Predictable? A Layer–Time Lens for Diffusion Editors

Anonymous submission

Abstract

Between reading an edit instruction and painting pixels, what does a diffusion-transformer (DiT) image editor represent? We build a layer–time lens that decodes intermediate residuals through the model’s frozen output head and pairs readouts with linear probes and activation interventions (ablation, transplant, steering). On fixed underspecified recoloring prompts, the *realized* outcome is predictable from early layers while the frozen head still reads noise (car color: 0.667 balanced accuracy vs. chance 0.333, $p=0.002$). That target-image code is *translated late* into the velocity field in a sharp, timestep-invariant band (layers 52–54 of 60). The boundary persists across tested edit families, an independent fine-tune, and a four-step distilled checkpoint; held-out early best-of-five selection validates the readout on a second architecture. Transplant establishes sufficiency; steering shows writable hue requires a spatial carrier. Truncating classifier-free guidance after step two saves 45% of forward passes on tested edits.

Introduction

Between reading “turn the car red” and painting it, what does a diffusion image editor actually do? The mechanics of diffusion suggest a gradual-development story, each of sixty blocks and twenty denoising steps contributing its small share – but this paper measures that story and finds it wrong. In the model we study, the realized edit is *predictable early* from an internal target-image code, and only *translated late*, in the last few blocks, into the flow field that does the painting.

The logit lens (nostalgebraist 2020) and tuned lens (Belrose et al. 2023) track a language model’s next-token prediction by projecting intermediate residuals through its final unembedding. DiT image editors (Peebles and Xie 2023; Qwen Team 2025) have no analogous vocabulary: a patch hidden state is not an output distribution, so asking what a layer is drawing has required running the model to the end.

We build a layer–time lens for a production DiT editor, Qwen-Image-Edit-2511 (60 blocks), by decoding intermediate residual streams through the model’s own frozen output head, and – crucially – pair passive readings with linear probes and controlled activation interventions (ablation,

transplant, steering) that test what those activations can install or write.

Naively porting the logit-lens recipe reproduces the familiar mid-stack “nothing happens yet” appearance – and that appearance is itself misleading. With source and an underspecified recoloring instruction fixed across 48 seeds, a generation-level probe predicts the color ultimately chosen from layer 6, where the frozen head still reads noise. The outcome is held in a basis the head cannot parse until a narrow band near the top translates it into the velocity field that paints the result.

The main contributions of this work are as follows:

- **An instrument.** A layer–time instrument for a DiT image editor, combining a training-free frozen-head lens with tuned and differential variants, paired with linear probes and three intervention primitives (ablate, transplant, steer) with stated write semantics, under a falsification-first protocol – pre-registered predictions, per-image visual verdicts, and bit-exact gates.
- **A mechanism: the editor thinks with an internal image.** The realized edit is predictable early under a fixed underspecified instruction, carried as a target-image code, and translated into the velocity code only in a sharp, timestep-invariant band (layers 52–54, transition width one layer) that is consistent across the edit families tested and survives an independent fine-tune and a 4-step distillation. Transplant establishes sufficiency; steering and a rank ladder show that readable attributes require a spatial carrier and a variable-dependent write subspace.
- **Validation and a lever.** Held-out early best-of-five selection validates the readout on Qwen and an independent architecture, with about 68% measured compute reduction. Separately, truncating classifier-free guidance after step 2 saves 45% of forward passes across tested edit families, suggesting CFG’s non-redundant role here is scope resolution rather than outcome selection.

This is exploratory, single-model-family work; every verdict was made on the images themselves, and we are explicit about where the lens is unreliable.

Related Work

Logit and tuned lenses. The logit lens (nostalgebraist 2020) projects intermediate residual streams through a lan-

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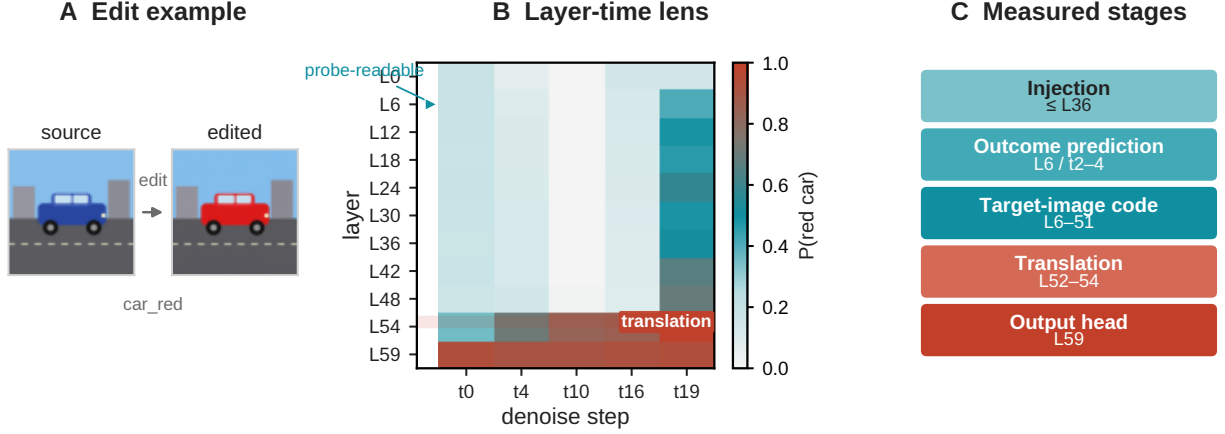


Figure 1: **Predictable early, translated late.** **A:** source and realized edit. **B:** frozen-head $P(\text{red car})$ across layer and step; the outcome is probe-readable at L6, but translation into the head-readable velocity convention occurs at L52–54. **C:** the five measured stages used throughout the paper.

guage model’s frozen unembedding; the tuned lens (Belrose et al. 2023) fits a small per-layer affine map instead. Both were built for autoregressive text models, where the unembedding gives every layer a shared, well-defined target vocabulary; a DiT editor has no such vocabulary; a patch’s hidden state is not a distribution over anything until the whole network has run. Recent Jacobian-style grids likewise relate internal activations to output units along depth; we transpose that idea to layer–time readout cells (§) with decode-then-CLIP and linear probes on a frozen editor. We port both lens families to this setting – staying on the frozen-head side for honesty and crossing to the learned side only to render the internal image – with the linear representation hypothesis (Park, Choe, and Veitch 2024) licensing the informative disagreement between a frozen head and a probe.

Instruction-based diffusion image editing. Two lines established how a diffusion model can be made to follow an edit instruction at all. InstructPix2Pix (Brooks, Holynski, and Efros 2023) trains directly on synthesized instruction-image pairs so the model learns the edit end to end; Prompt-to-Prompt (Hertz et al. 2023) instead manipulates the model’s cross-attention maps at inference time to inject the edit while preserving the source’s structure, training-free. Both contribute a *mechanism for producing* the edit; neither asks what representation the network forms once an instruction is given, or where in the forward pass it becomes the pixels that change. Our lens targets that open question on exactly such a production editor – not a new editing mechanism, but a reading of the one it already has.

Thinking with images, and diffusion interpretability. A growing line has multimodal models reason *with* explicit intermediate images by externalizing a visual scratchpad (Hu et al. 2024; Wu et al. 2024; Li et al. 2025). We supply the mechanistic complement: an image that exists *latently* inside a generative editor before any output pixel is produced, with

no explicit scratchpad or tool call involved. The Diffusion Lens (Toker et al. 2024) decodes a text-to-image model’s *text encoder*; DiffLens and ICM (Shi et al. 2025; Zaleska et al. 2026) probe and steer internal features, including choices absent from an underspecified prompt; DreamReader unifies extraction, patching, steering, and stitching for text-to-image U-Nets (Sykora et al. 2026); and sparse autoencoders expose causal generation features (Surkov et al. 2025). Breaking the Lock-in finds early cross-seed convergence in text-to-image sampling and intervenes to increase diversity (Zhu et al. 2026). Probe-Select (Guo, Wei, and Jing 2026) predicts final quality from early activations; ELECT (Kim et al. 2025) scores early latent background consistency; and ADE-CoT (Qu et al. 2026) adds adaptive budgets and stopping at scale. We instead predict *which outcome* a fixed underspecified edit realizes and use selection only as a held-out behavioral test, while mapping how an image-editing DiT’s realized outcome changes code across depth; we do not infer irreversible lock-in. We decode an editing DiT’s own denoising stack as an image across both layer and time, building on the flow-matching and rectified-flow formulation (Lipman et al. 2023; Liu, Gong, and Liu 2023) and classifier-free guidance (Ho and Salimans 2022) that Qwen-Image-Edit-2511 (Qwen Team 2025; Peebles and Xie 2023) uses.

Causal interpretability and editing. Causal tracing / ROME (Meng et al. 2022) patches activations between runs, the move behind our transplant; Heimersheim and Nanda (2024) catalogue patching pitfalls that motivate our cautious interpretation of ablation. Recent analyses further show that detector-friendly diffusion features can produce artifacts when steered off distribution (Pegoraro, Johnson, and Nixon 2026), and that patching with multiple mediators cannot by itself isolate a unique causal pathway (Zhang, Hegde, and Zhang 2026). Accordingly, transplant is used only as a sufficiency test and steering as a bounded write test. Finally, activation addition (Turner et al. 2023) is the steering prim-

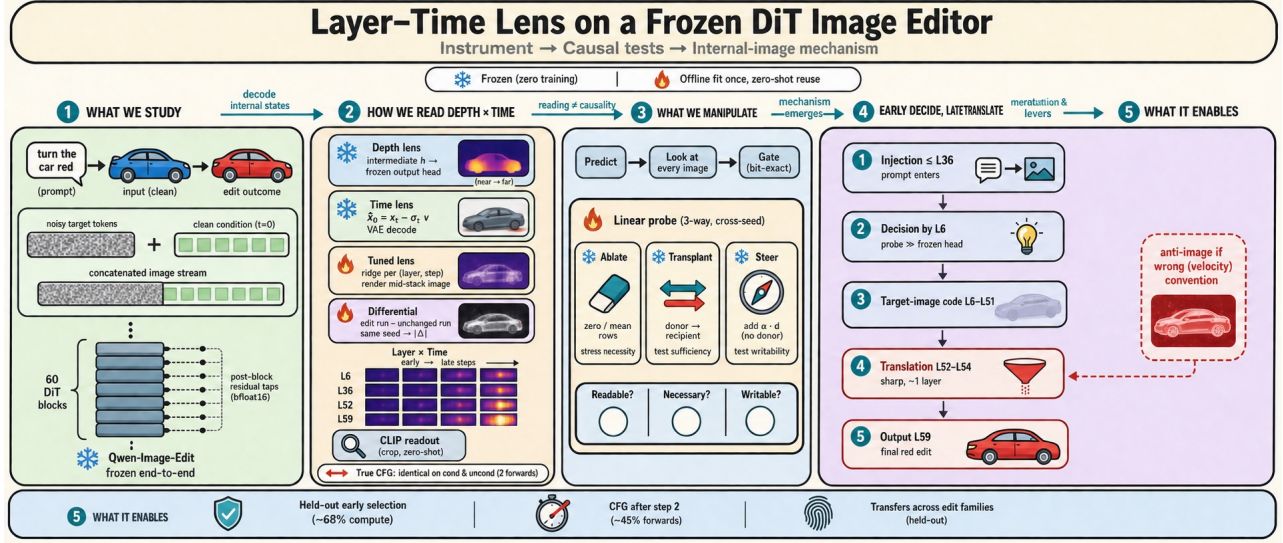


Figure 2: **Framework overview.** Frozen Qwen-Image-Edit-2511 on a concatenated noisy-target / clean-condition image stream with post-block residual hooks (`bfloat16`). *Read*: depth, time, tuned, and differential lenses decode previews, then crop-level CLIP scoring on matched true-CFG passes. *Test*: linear probes plus ablate, transplant, and steer interventions under predict–look–gate protocol. *Mechanism* (schematic): early decision and target-image code, late L52–L54 translation into the velocity field; bottom: held-out validation and training-free levers discussed in text.

itive we instantiate with a probe direction. All three were built for language models with a token vocabulary at every patched site; we port them to a continuous, spatial image stream, where sufficiency must be checked against a rendered image rather than a next-token distribution. Concept Lancet (Luo et al. 2025) already uses compositional representation transplant for diffusion editing; our transplant is therefore an interventional sufficiency check within a layer–time map, not a standalone editing-method claim. Our semantic readout uses CLIP (Radford et al. 2021) for lack of a better zero-shot alternative.

Proposed Method: A Layer–Time Lens and Intervention Probes

Figure 2 summarizes the full pipeline; here we specify each component. We study Qwen-Image-Edit-2511, a 60-block DiT editor whose image stream concatenates noisy target tokens with always-clean condition tokens. All quantities below are read from the *post-block residual* of selected blocks via forward hooks, captured in the model’s native `bfloat16`: image-stream activations reach `absmax` $\sim 3 \times 10^9$ even at layer 0, so a cast to `float16` silently overflows most elements to `inf` (bf16 capture is a lossless copy, not a downcast).

Notation and layer–time readout. Let $h_{\ell,t}^{(i)} \in \mathbb{R}^d$ denote the post-block residual after block $\ell \in \{0, \dots, 59\}$ at denoising step t , for image-stream row i (target patch, condition patch, or a hooked subset); $h_{\ell,t}$ stacks the rows we capture. Logit- and Jacobian-style lenses in language models align internal states with output units along depth and token position; a DiT patch has no vocabulary, so we define a *layer–time lens*

as a map from (ℓ, t) to a score on a fixed grid:

$$\mathcal{M}_{\ell,t} = \mathcal{R}(\mathcal{D}(\mathcal{L}_{\ell,t}(h_{\ell,t}))), \quad (1)$$

where $\mathcal{L}_{\ell,t}$ is one of the lens operators below, $\mathcal{D}$ is VAE decode when $\mathcal{L}$ returns a latent velocity, and $\mathcal{R}$ crops the edited region and applies CLIP (Radford et al. 2021) over labels $\{y_k\}$,

$$[\mathcal{M}_{\ell,t}]_k = \text{softmax}_k(\text{CLIP}(y_k, \text{crop}(\mathcal{D}(\cdot))))). \quad (2)$$

Figures 1 and 2 instantiate (1)–(2): the image-editor analogue of lexical alignment grids in logit/Jacobian-lens visualizations, not a full $\partial y / \partial h$ (a donor-scalar Jacobian basis is tested in the supplement and does not coincide with probe or write directions).

Lens operators. The *time lens* uses the terminal head output at step t :

$$\hat{x}_0(x_t, t) = x_t - \sigma_t v_\theta(x_t, t), \quad (3)$$

decoded through the VAE—what the model currently believes it will output. The *depth lens* asks the same question at intermediate depth by re-invoking the model’s frozen output head (non-affine `AdaLayerNormContinuous` + linear `proj_out`, bit-exact whether invoked at layer 59 or earlier):

$$v_{\ell,t} = \text{Head}(h_{\ell,t}), \quad \hat{x}_{0,\ell,t} = x_t - \sigma_t v_{\ell,t}, \quad (4)$$

then $\mathcal{D}(\hat{x}_{0,\ell,t})$. Because `norm_out` has no learned scale, rescaling $h_{\ell,t}$ to the terminal norm is a no-op (raw vs. rescaled agree to `max_rel_diff` = 0); we use the raw head everywhere with a standing self-consistency gate on every grid. The *tuned lens* (Belrose et al. 2023) fits a per- (ℓ, t) ridge map

from $d=3072$ hidden dimensions to the 64-d conditional-pass velocity, one dictionary per cell,

$$\hat{v}_{\ell,t} = W_{\ell,t}h_{\ell,t} + b_{\ell,t}, \quad W_{\ell,t} = V^\top H(H^\top H + \lambda I)^{-1}, \quad (5)$$

after token-wise centering; λ is chosen per cell on a held-out 10% *token* split, distinct from zero-shot tests that hold out an entire edit family and source image (hyperparameters in the supplement). The *differential lens* differences edit and control runs that share a seed (step-0 latents are then bit-identical):

$$D_{\ell,t} = |\mathcal{D}(\mathcal{L}_{\ell,t}(h_{\ell,t}^{\text{edit}})) - \mathcal{D}(\mathcal{L}_{\ell,t}(h_{\ell,t}^{\text{ctrl}}))|, \quad (6)$$

isolating the internal edit plan when the unchanged prompt is a faithful no-op; a control-fidelity gate on correlation with the source flags cases where it is not.

Linear probes. Neither (2) nor a lens image is itself a causal claim. We additionally fit a lightweight probe $p(y | h_{\ell,t}) = \text{softmax}(Wh_{\ell,t})$ on hidden states (cross-seed evaluation), reading the same (ℓ, t) grid in label space rather than pixel space—the basis mismatch against the frozen head in (4) is central to our mechanism results.

Intervention probes. Three forward-hook overwrites of the post-block residual test distinct properties, applied identically to both CFG passes:

$$\begin{aligned} h &\leftarrow \mathbf{0} \text{ or } \bar{h} && \text{(ablate: necessity stress test),} \\ h^{(\text{rec})} &\leftarrow h^{(\text{don})} && \text{(transplant: sufficiency),} \\ h &\leftarrow h + \alpha d && \text{(steer: linear writability),} \end{aligned} \quad (7)$$

with α in units of per-row RMS and d a probe direction; an empty ablation spec reproduces the baseline bit-for-bit. Repeated ablation can drive activations off-manifold, so we treat it only as a coarse stress test.

Protocol: predict, look, gate. Three rules make the *reasoning* reproducible, not just the runs. **Predict, then run:** every intervention was pre-registered; refutations are reported as prominently as confirmations, and several headline results are pre-registered predictions that failed informatively. **Look at every image:** no verdict rests on CLIP alone – every run carries a per-image visual verdict recorded before the aggregates were read, and where the two disagree the image wins. **Gate the instrument:** the empty-ablation and raw-vs-rescaled checks are standing regressions re-run per experiment. Porting the method to another editor needs only a native-dtype residual hook, the frozen-head gate, and region-restricted probes – the recipe executed unchanged in the cross-model replication below.

Experiments

Experimental Setup

We evaluate on Qwen-Image-Edit-2511, a 60-block DiT editor, on edit batteries spanning five categories – color, background, style, removal, and addition – over natural photographs and one flat-art source (10 cases for the time-lens saturation sweep and the preview lever, 20 for the IoU-based

onset and CFG-truncation checks); every quantitative number is paired with a per-image visual verdict recorded before the aggregate is read, per the predict-look-gate protocol above. Three readouts do the reporting: the frozen output head’s decode, scored zero-shot with CLIP against a small label bank; cross-seed probes, including a 48-seed fixed-prompt outcome probe whose independent unit is one generation; and, for the tuned lens, held-out R^2 against the true conditional-pass velocity. Two further checkpoints sharing Qwen-Image-Edit-2511’s transformer classes – FireRed-Image-Edit-1.1, an independent fine-tune, and Rapid-AIO, a 4-step distilled merge – let us ask whether a finding persists beyond one training run (tested below).

The Realized Edit Is Predictable Early

Question. Does the edit unfold gradually, or is the final outcome readable before the frozen head agrees?

In time. If the edit surfaced progressively, the readout would saturate late and at roughly the same rate. Running the time lens over a battery of 10 cases across 5 edit categories, the step at which $P(\text{target})$ first reaches within 0.05 of its final value is instead *edit-dependent* and mostly *immediate*: color, background, and style edits reach the criterion at step 0–2, and only a wholesale new-object insertion (*add_boat*) does so as late as step 7. A finer IoU-mask-based onset measurement over 20 cases shows the split is categorical, not continuous: 18 of 20 – spanning final-edit-area fraction 0.003 to 0.99 – onset at step 0 regardless of how large or novel-looking the finished edit is, and no continuous proxy predicts the delay (edge-novelty $\rho=0.24$, $p=0.30$; area $\rho=-0.44$, running the *wrong* way). Only the two edits that insert genuinely novel structure with no existing pixels to build on – *add_boat* and *add_runner* – start late.

In depth, the naive lens says “late”. The frozen depth lens tells a different-looking story. Decoding *car_red*’s intermediate layers through the model’s own head, the edit becomes head-readable only in a sharp snap between layers 48 and 54 of 60 – exactly the familiar mid-stack “nothing happens yet” appearance of an LLM logit lens.

But the realized outcome is predictable 40 layers earlier.

An explicit-prompt probe first reveals the basis mismatch: predicting which of three colors was requested reaches 0.883 accuracy at **layer 6**, step 4, where the frozen head reads only $P(\text{red car})=0.111$ (Figure 3). To separate outcome prediction from reading a color word, we hold both source and the underspecified prompt “change the car to a different color” fixed across 48 new seeds and label each final image post hoc. One mean-pooled vector per generation predicts the realized color at L6/step 4 with 0.667 balanced accuracy (chance 0.333; $p=0.002$). Thus the outcome, not merely the instruction, is linearly predictable early; this does not claim irreversible commitment.

Prediction generalizes and supports selection. Table 1 summarizes continuous-hue replications and held-out best-of-five tests. Natural-photo mug and tree edits replicate while the heterochromatic-eyes task is null ($p=0.796$); the independent 40-block Boogu editor also carries an early hue readout,

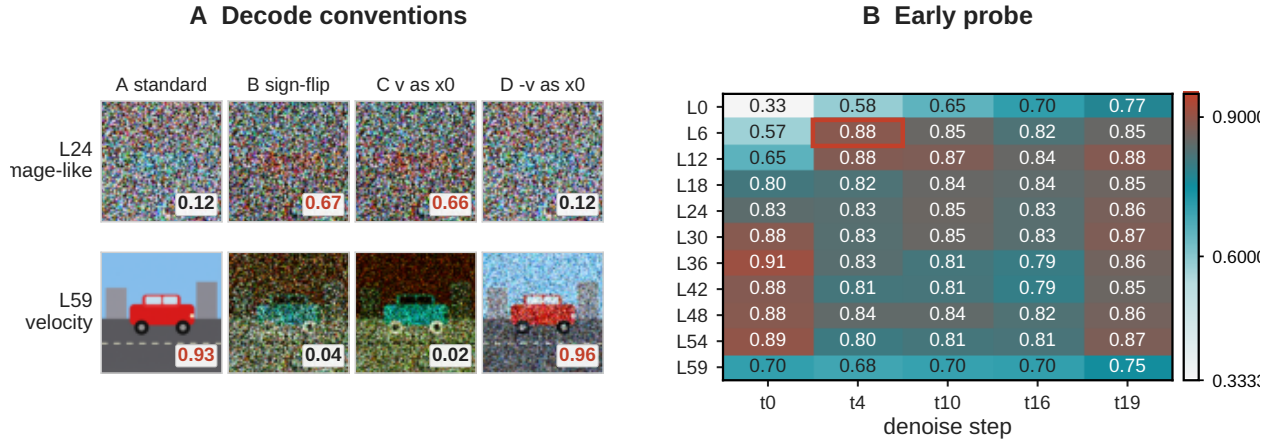


Figure 3: **Decode convention flips; the probe reads early.** **A:** four *car_red* decodes reverse convention between L24 and L59. **B:** the 3-way probe reaches 0.883 at L6/t4 while the frozen head still reads noise (chance 0.33).

Table 1: Early prediction and held-out selection. Q/B denote Qwen/Booggu; parentheses give chance or random-selection baselines.

Setting	Readout	Metric	Result	Evidence
Car fixed	Q L6/t4	Bal. acc.↑	.667 (.333)	$p=.002$
Car hue	Q L6/t2	MAE↓	22.3→8.3°	$R^2=.815$
Tree hue	Q L6	MAE↓	55.9→11.4°	$p=.005$
Mug hue	Q L6	MAE↓	17.0→10.7°	$p=.005$
Boogu hue	B L4/t2	MAE↓	39.9→6.9°	$p=.005$
Boogu select	B L4 best-5	Success↑	8/12 (20%)	$p=4.1\times 10^{-5}$
Tree select	Q L6/t2	MAE↓	95.2→46.5°	$p=3.7\times 10^{-7}$
Car select	Q best-5	MAE↓	25.7→11.1°	$p=.0086$

although its terminal layer is stronger. Selection succeeds on Boogu and two Qwen target tasks. Actual interrupted captures plus winner reruns save about 68% versus full best-of-five, but only select the closest available candidate.

Where the instruction enters. A complementary text-stream ablation locates where the prompt is *consumed*: zeroing the model’s text tokens in only the first 11 blocks garbles the whole scene, while zeroing them after layer 36 dents the edit ($P(\text{red})=0.993$). The instruction is fused into the image stream early; for this color edit, later text tokens are dispensable by layer 36 – the “injection” stage of the circuit in Figure 1, whose full text-band table is in the supplement. Together with the probe, this gives the paper’s five-stage picture: injection ($\leq L36$), early outcome prediction, a target-image code, translation (L52–54), and the output head. **Takeaway.** Time onset is mostly immediate; the frozen depth lens looks late, but generation-level probes read the realized outcome from L6.

The Edit Is Translated Late: Two Codes

Question. If probes read the outcome early, what mid-stack code is converted at L52–54? Three measurements answer

this: at an intermediate layer the frozen head yields a per-token vector v_ℓ we normally treat as a velocity, forming $\hat{x}_0 = x_t - \sigma_t v_\ell$ (variant A). Decoding v_ℓ instead *directly*, as if it were already a clean latent (variant C), inverts the story: for *car_red* at step 4, at L24 the standard decode A scores $P(\text{red})=0.12$ (still wrong-colored) while the direct decode C already scores 0.66; at L59 this reverses (A = 0.93, C = 0.02). The mid-stack is not blank – it encodes the target content in an *image-like* convention, and a narrow band near the top reconciles it to the velocity convention the head consumes. (This is a whole-vector reweighting, not a literal sign flip: essentially no individual channel of v_ℓ has a negative slope against v_{59} , mean 0.003.)

The conversion is a sharp, timestep-invariant band. A dense every-layer sweep from L44 to L59 (Figure 4) locates the A/C crossover precisely: transition midpoint L53.6 at step 16 (width one layer) and L52.7 at step 4, the two differing by under one layer – the boundary’s *location* is timestep-invariant. Crucially, the cross-seed probe accuracy is close to *flat* (0.77–0.84) across every layer from L44 through L58 and drops only at the terminal layer L59 (0.68–0.70). So the translation that changes what the head can read (L52–54) and the loss of linear separability (L59) are two *distinct* events sharing a neighborhood: the target-image code survives the translation intact and is degraded only by the final, velocity-specific projection.

A tuned lens renders the internal image, layer by layer.

To confirm the mid-stack code is a genuine *image* rather than a probe artifact, we fit a per-(layer, timestep) tuned lens, as described above, on *remove/background/addition* edits and evaluate it *zero-shot* on the held-out *car_red* color family. Held-out R^2 rises with depth from 0.73–0.89 at L6 to 0.998 at L59, whose frozen-head decode reproduces the true velocity bit-for-bit. Every tested depth down to L6 yields a coherent scene, refuting a pre-registered readability boundary: L52–54 marks the frozen head’s projection, not the representation. At step 0 the pixels are pure noise, yet Figure 5 shows the

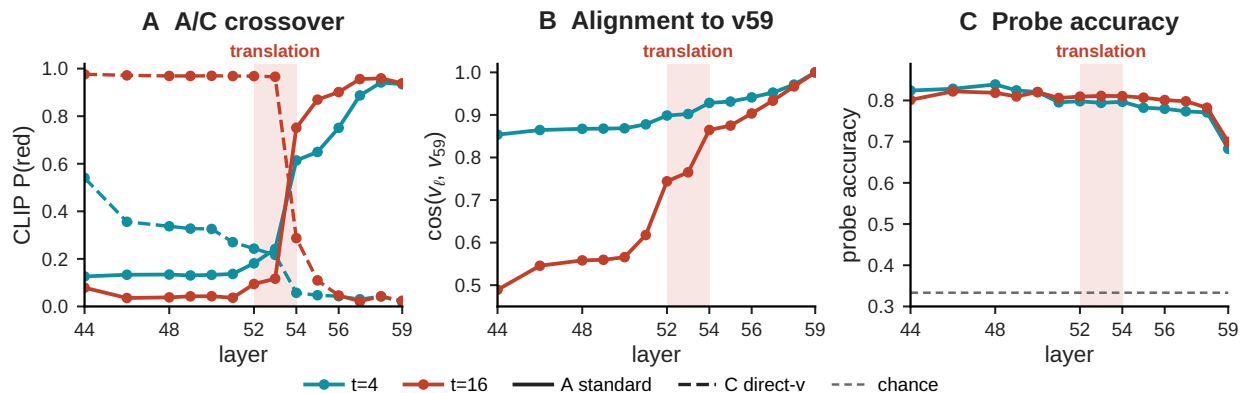


Figure 4: **Translation band L52–54.** Dense L44–L59 sweep on *car_red*. **A–B:** decode crossover and velocity alignment change in the same band at $t=4, 16$. **C:** probe accuracy stays flat through L58 and drops only at L59.

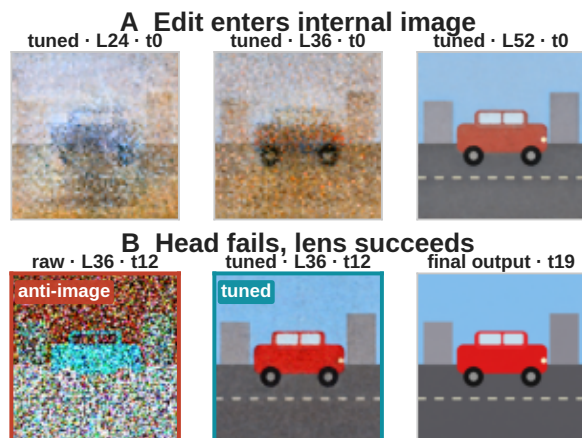


Figure 5: **Internal image vs. frozen-head ghost.** **A:** the zero-shot tuned lens shows the edit enter a coherent internal image at $t=0$. **B:** for the same L36/ t_{12} state, the frozen head yields an anti-image while the tuned lens recovers the edited scene.

blue car at L24, orange-red at L36, and fully red at L52. The dictionaries transfer without retraining to six further held-out edits; *add_runner* also reproduces delayed onset internally.

The anti-image, mechanized. This also explains a striking artifact: the mid-stack *frozen-head* decode at later steps is a color-inverted ghost (a cyan car under a red-brown sky) – the image-like mid-stack code forced through a head built to consume a velocity, since $\hat{x}_0 = x_t - \sigma v$ flips the sign of an image-like quantity. The tuned lens at the same cell is already coherent. **Takeaway.** Mid-stack states are image-like; the L52–54 band reconciles them to the velocity convention the head consumes.

Transplantable, Writable, and Carrier-Bound

Reading a lens shows only that a representation is *consistent with* an edit, not what it can install. Whole-token ablation

across many layers drives the edited region off-manifold, so we use it only as a coarse stress test rather than evidence for fine-grained necessity.

Transplant establishes sufficiency. Transplanting a *car_red* donor’s car-region hidden states into a *car_green* recipient (which alone scores $P(\text{red})=0.011$) flips the output to red from *any* of four donor bands, including **L48–59** (0.940). The donor representation is therefore sufficient to install the edit. A content control confirms specificity: splicing the same donor’s *background*-region tokens (right band, wrong content) yields a garbled “blue car” (0.991), not red.

Readability does not imply unconstrained writability. A single stored *direction* – the probe’s d_{red} , no donor pass – added to the recipient’s car rows gives a sharp sigmoidal dose-response (0.024 at $\alpha=0.5$, 0.977 at $\alpha=1$, saturating to $\alpha=8$); a random direction of equal norm is inert (0.017), and the negated direction overshoots to blue (0.945), reproducing the anti-image under an active intervention. But writability is bounded: at effective doses the steered region is not a recolored car but a flat red patch that *replaces* the car’s geometry – the direction installs *which* hue, not *where to stop*. Supplying that missing “where” with the edited object’s own silhouette mask turns every previously-failing dose into a clean in-place recolor. And conjuring an *object* outruns any single direction: the identity probe direction is inert (the read and write subspaces differ), while a rank ladder over the donor–recipient difference resolves identity coarse-to-fine – category at rank 1, a coherent instance by rank 4, donor-faithful by rank 64 (2% of the model dimension). The read and write subspaces differ, and a writable attribute still requires a spatial carrier.

A Preview Lever: The Differential Lens at 15% NFE

The differential lens (*Proposed Method*) isolates the internal edit plan from the reconstruction. It first settles a caveat left open above: token-level R^2 *inside* the edit region is 0.88–0.97, matching the full-image value, so the tuned dictionary

Table 2: Cross-checkpoint translation and early readout. Transfer is Qwen→FireRed / FireRed→Qwen; L12/t4 direction cosine is .833.

Checkpoint	A/C band	Probe: early → L59	Transfer
Qwen	L52–54	.883 @ L6/t4 → .68–.70	source
FireRed	L53–56	.852 @ L6/t4 → .69–.70	.840/.917
Rapid-AIO	L52–54	–	–

ies carry the edit-specific signal itself, not incidental scene detail.

Content is early; separation is gradual. A pre-registered prediction that D would already localize tightly at every step is refuted, informatively. The edit *content* is present at step 0 (the tuned lens already shows red eyes, a black trunk), but the *differential* at step 0 is a diffuse whole-frame field – the edit prompt perturbs the entire velocity relative to the no-op, not just the edited region – and condenses onto the edit’s footprint only gradually (localization of D ’s top mass at L52 climbs monotonically across steps, e.g. $0.21 \rightarrow 0.47 \rightarrow 0.67 \rightarrow 0.83$ for a cat’s-eyes recolor). *Present* (content in the internal image) and *separated* (the differential has found where it lives) are different quantities. The no-op control also caught one flat-art scene that *cannot* “do nothing” (it renders a transparency checkerboard, and is discarded), motivating the control-fidelity gate applied before any differential number is trusted.

A preview lever. Because the differential carries the edit’s footprint, we ask whether it does so early enough to be useful. At step 2 – the third of 20, $\approx 15\%$ of the forward passes – across a 10-case, five-family battery of natural photographs (all passing the control-fidelity gate, $r \in [0.86, 0.99]$), the median IoU between $D_{L52,t2}$ ’s top mass and the true edit region is 0.33, $6.6\times$ chance, and all 10 cases sharpen from step 0 to step 2 (figure in the supplement): an object slated for removal lights up as a ghost, a keep-vs-replace background segmentation emerges unsupervised, and a tiny recolor glows at its exact site. Because the no-op control is *independent of the edit prompt*, it is computed once per source image and cached; previewing a *candidate* edit then costs three forward passes plus one frozen-dictionary decode, not a full 20-step run – a training-free way to triage many prompts against one image before running any to completion.

Generalization Across Checkpoints

Most mechanistic findings so far concern one architecture. We re-ran the depth-lens crossover and cross-seed probe on the two further checkpoints from *Setup* (FireRed-Image-Edit-1.1 and Rapid-AIO). **The translation boundary and explicit-prompt early readability replicate** (Table 2). FireRed has the same flat-then-dip probe shape, and Rapid-AIO preserves the boundary under a four-step schedule. The readable direction also transfers between Qwen and FireRed, rather than merely recurring at the same layer index. The full tuned lens ports to FireRed as well (deep-layer R^2 0.97–0.999; anti-image reproduced).

A Training-Free Lever: Truncating Classifier-Free Guidance

The layer–time map suggests a simple existence proof: classifier-free guidance may not need both forward passes at all 20 steps. Running the unconditional pass only for the first two steps and the conditional pass alone thereafter leaves all four tested edit families visually intact at a **45% reduction in forward passes** (a 20-case battery confirms it, mean $1.77\times$ speedup). Dropping CFG entirely suggests where its non-redundant role lies in these cases: recolor, removal, and addition still land from the conditional pass alone, but background swap composites the new scene *around* un-erased source structure ($P(\text{beach})=0.45$ vs. ≥ 0.996) – CFG’s non-redundant contribution here is consistent with resolving *scope*, not selecting edit content. The same layer–time structure survives $7.7\times$ step distillation (20/20 cases land), so distillation compresses the sampling trajectory but not the structure the lens exposes.

Routing conservation explains coherent transplant. A RESTORE-inspired five-seed audit (Cho et al. 2026) finds destructive replacement more routing-distorting than transplant/red steering in 5/5 seeds (contrast 0.0356, exact $p=0.03125$). Matched-donor TV is 0.0031 with geometry 5/5; background-donor TV is 0.0677 with geometry 0/5. Red steering writes red 5/5 but preserves geometry 0/5, matching the detection–intervention gap (Pegoraro, Johnson, and Nixon 2026); this supports tested coherence only.

Limitations

Translation/intervention evidence uses Qwen plus two same-lineage checkpoints; Boogu supports only outcome prediction/selection. The fixed-prompt probe has one synthetic source; natural-image follow-ups give two replications and one unsuitable eye case. Selection has 12 groups per task and cannot create absent targets. One-way L6 hue transfer (mug→tree $p=0.002$; reverse $p=0.968$) precludes an object-invariant subspace. Because CLIP can prefer a flat patch to a true recolor, quantitative claims carry pre-registered visual verdicts. We claim tested outcome predictability, transplant sufficiency, and carrier-bounded writability, not irreversible commitment or necessity.

Conclusion

Across the tested editors, early probes predict realized edits; in the Qwen lineage, an image-like code is translated to velocity at L52–54. Lenses and interventions bound readability, sufficiency, and carrier-bounded writability, while held-out selection and CFG truncation demonstrate practical use. We do not claim irreversible commitment, necessity, or architecture-wide universality.

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